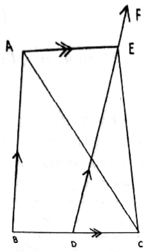
AMTI (NMTC) - 2011

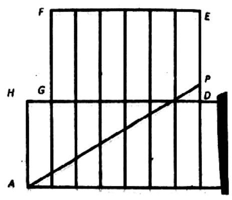
GAUSS CONTEST - FINAL - PRIMARY LEVEL

1. Pustak Keeda of standard six bought a book. On the first day he read one fifth of the number of pages of the book plus 12 pages. On the second day he read one fourth of the remaining pages plus 15 pages and on the third day he read one third of the remaining pages plus 20 pages. The fourth day which is the final day he read the remaining 60 pages of the book and completed reading. Find the total number of pages in the book and the number of pages read on each day.
2. In the adjoining figure $\angle A$ is equal to an angle of an equilateral triangle. DEF is parallel to AB and AE parallel to BC. $\angle CEF = 170^0$ and $\angle ACE = \angle B + 10^0$. Find the angles of the triangle and $\angle CAE$.



3. $p = 1 + 2^1 + 2^2 + 2^3 + ... + 2^n$ where p is a prime number and n is a natural number. Find all such prime numbers $p < 100$ and the corresponding natural number n. For each (p, n) find $N = p \times 2^n$ and find the sum of all divisors of N.
4. The sequence 8, 24, 48, 80, 120, ... consists of positive multiples of 8, each of which is one less than a perfect square. Find the 2011th term. Divide it by 2012 and find the quotient.
5. Each letter of the following words is a positive integer. The letters have the same value wherever they occur. The numerical values given for each word is the product of the corresponding numbers of the letters appearing in the word.
6. (a) The length of the sides of a triangle are three consecutive odd numbers. The shortest side is 20 % of the perimeter. What percentage of the perimeter is the largest side ?

   (b) The sides of the triangle are three consecutive even numbers and the biggest side is $44\frac{4}{9}\%$ of the perimeter. What percentage of the perimeter is the shortest side ?
7. In the figure all the 14 rectangles are equal in size. The dimensions of each rectangle are 2 unit x 5 units. P is a point on ED. AP divides the octagon ABCDEFGH into two equal parts. Find the length of AP. (Hint : Area of a triangle = $\frac{1}{2}$ base x height).



8. In rectangle ABCD, the length is twice the breadth. In the square each side is equal to one unit more than the breadth of the rectangle. In the triangle LMN, the altitude is one unit less than the breadth of the rectangle. Area of the rectangle is 18 square units. The sum of the areas of the rectangle and the square is equal to the area of the triangle. What is the base of the triangle and the areas of the square and the triangle.